Sahaj Bhadja

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Summary

Associate Software Engineer with 2+ years of experience building **Python analytics systems**, **SQL data pipelines**, scalable applications, and **automated testing workflows** over large datasets.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant analytics platform supporting 500+ users.
- Achieved 99%+ reliability across 5K+ monthly jobs through **incremental data pipelines**, implementing validation, reconciliation, error handling, cloud storage, CI/CD releases, and stable production observability reliably at scale.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python analytics platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Maintained 90%+ consistency across 10K+ submissions through **structured model evaluation**, persistent results, fallback paths, and reviewer controls supporting reproducible analytical workflows and dependable production decisions reliably at scale.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **quantitative modeling pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and reproducible evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through a **spatial-temporal model** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex simulation scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Increased pipeline efficiency by 68% through **AWS Glue and S3 ETL**, transforming JSON, CSV, and Parquet inputs into normalized downstream datasets while reducing storage costs by 15% reliably.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst - Research Fellowship | *India*
- Achieved 96%+ classification accuracy across 650K+ URLs by comparing **Random Forest, CNN, RNN, LSTM, Bi-LSTM, and GRU models** within a reproducible analytical pipeline for malicious-pattern detection reliably.
- Engineered 21 behavioral indicators covering URL length, IP presence, and suspicious keywords, translating noisy inputs into **classification features** that exposed phishing and malware patterns reliably at scale.

Skills

Languages: Python, Java, C++, SQL, TypeScript, JavaScript, Bash
Analytics & Modeling: Data Analysis, Statistics, Classification, Random Forest, CNN, RNN, LSTM, GRU, Feature Engineering, Model Evaluation
Data Engineering: PostgreSQL, MySQL, AWS RDS, AWS Glue, AWS S3, ETL, Data Pipelines, Data Validation, Schema Design, Query Optimization
Applications & Cloud: REST APIs, FastAPI, Flask, React, Node.js, AWS, Docker, Kubernetes, Linux
Quality & Delivery: Object-Oriented Programming, Data Structures, Algorithms, Unit Testing, Integration Testing, Automated Testing, PyTest, Mockito, Git, GitHub, Jenkins, CI/CD, Debugging, Observability

Projects

Healthcare Claims Analytics Platform - Design Concept
- Designed a **healthcare claims analytics platform** concept ingesting large clinical and financial datasets, validating records, identifying payment anomalies, prioritizing audit cases, and presenting traceable findings through reviewer dashboards.

SQL Audit Rules Engine - Design Concept
- Designed a **SQL audit-rules engine** concept with versioned validation logic, exception queues, query-performance monitoring, regression datasets, approval workflows, evidence reports, and repeatable release checks for analytical applications.

HMIS - Hospital Management Information System | Live
- Built a **TypeScript healthcare platform** with ABDM APIs, consent-based patient-history access, appointment workflows, Twilio alerts, and Sentry monitoring, reducing administrative workload 30% while supporting 100+ daily appointments.